

## WMO: 786370

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |    |      |     |    |      | Coldest Month WS/MCDB |      |    |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-----|-------------------------------|----|------|-----|----|------|-----------------------|------|----|------|-----------------------|------|-----|
|               |            |     | 99.6%                         |    |      | 99% |    |      | 0.4%                  |      | 1% |      |                       |      |     |
|               | 99.6%      | 99% | DP                            | HR | MCDB | DP  | HR | MCDB | WS                    | MCDB | WS | MCDB | MCWS                  | PCWD |     |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |    |      |    |      | Evaporation WB/MCDB |      |    |      |    |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|----|------|----|------|---------------------|------|----|------|----|------|----------------------|------|
|               |                        | 0.4%            |      | 1% |      | 2% |      | 0.4%                |      | 1% |      | 2% |      |                      |      |
|               |                        | DB              | MCWB | DB | MCWB | DB | MCWB | WB                  | MCDB | WB | MCDB | WB | MCDB | MCWS                 | PCWD |

|                                 |               |             |
|---------------------------------|---------------|-------------|
| Dehumidification DP/MCDB and HR | Enthalpy/MCDB | Enthalpy/DP |
|---------------------------------|---------------|-------------|

### Extreme Annual Design Conditions

| Extreme Annual WS |      |                    | Extreme Annual Temperature |     |            |     | n-Year Return Period Values of Extreme Temperature |     |            |     |     |     |  |  |
|-------------------|------|--------------------|----------------------------|-----|------------|-----|----------------------------------------------------|-----|------------|-----|-----|-----|--|--|
| Mean              |      | Standard Deviation | n=5 years                  |     | n=10 years |     | n=20 years                                         |     | n=50 years |     |     |     |  |  |
| 1%                | 2.5% | 5%                 | Min                        | Max | Min        | Max | Min                                                | Max | Min        | Max | Min | Max |  |  |

### Monthly Climatic Design Conditions

| Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| (d)    | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) |

|      |      |       |     |     |     |     |     |     |     |     |     |     |     |     |     |      |
|------|------|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| (14) | Wind | WSAvg | 3.5 | 3.7 | 3.6 | 3.8 | 3.8 | 3.6 | 3.5 | 3.9 | 3.4 | 3.1 | 3.0 | 3.0 | 3.0 | (14) |
|------|------|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|

|      |    |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|----|------|------|------|------|------|------|------|------|------|------|------|------|------|
| (19) | DB | 86.4 | 89.3 | 93.0 | 95.3 | 93.6 | 93.4 | 91.5 | 91.7 | 93.4 | 91.7 | 88.2 | 86.6 | (19) |
|------|----|------|------|------|------|------|------|------|------|------|------|------|------|------|

|      |    |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|----|------|------|------|------|------|------|------|------|------|------|------|------|------|
| (27) | WB | 78.6 | 80.0 | 82.0 | 83.5 | 83.6 | 83.5 | 82.4 | 82.7 | 82.6 | 82.3 | 80.4 | 80.2 | (27) |
|------|----|------|------|------|------|------|------|------|------|------|------|------|------|------|

|      |  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|--|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| (35) |  | MDBR | 12.4 | 13.6 | 14.9 | 15.0 | 14.0 | 13.3 | 12.9 | 13.6 | 13.7 | 12.5 | 12.3 | 12.0 | (35) |
|------|--|------|------|------|------|------|------|------|------|------|------|------|------|------|------|

|      |      |       |       |       |       |       |       |       |       |       |       |       |       |      |
|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
| (40) | taub | 0.382 | 0.395 | 0.429 | 0.510 | 0.559 | 0.474 | 0.487 | 0.459 | 0.437 | 0.426 | 0.401 | 0.403 | (40) |
|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|

|      |               |        |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|---------------|--------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| (44) | All-Sky Solar | BadAvg | 1161 | 1474 | 1707 | 1827 | 1700 | 1669 | 1708 | 1736 | 1670 | 1330 | 1151 | 1129 | (44) |
|------|---------------|--------|------|------|------|------|------|------|------|------|------|------|------|------|------|

|      |           |        |     |     |     |    |     |     |    |    |    |     |    |     |      |
|------|-----------|--------|-----|-----|-----|----|-----|-----|----|----|----|-----|----|-----|------|
| (45) | Radiation | RadStd | 134 | 128 | 131 | 91 | 112 | 121 | 72 | 90 | 72 | 146 | 80 | 109 | (45) |
|------|-----------|--------|-----|-----|-----|----|-----|-----|----|----|----|-----|----|-----|------|

| DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |